

Fast Ferment Test

From German brewing and more

The idea of the Fast Ferment Test is to completely ferment all fermentable sugars in a sample of wort. The attenuation that can then be determined is the limit of attenuation of that wort and it only depends on the wort composition that has been established by the brewing process (mainly mashing) [Narziss, 2005]. Lager yeasts are able to ferment some types of sugar (melibiose and raffinose) that ale yeasts can't or only partially ferment. But these sugars are either absent from brewer's wort or only present in trace amounts. When properly done, the final attenuation of the test is not affected by the yeast strain, yeast health, pitching rate, temperature and other fermentation parameters.

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How to perform the fast ferment test?

To perform the test you need a sample of wort and yeast. The amount of wort should be enough to perform a hydrometer reading later (6 to 8 oz (120 to 200 ml)). The amount of yeast should result in a pitching rate for that sample that is well above the pitching you would use in a beer (5-10x). Here is a quick and easy way to prepare the test when propagating yeast in a flask or large bottle (growler):

- decant the spent starter beer
- add fresh wort to resuspend the yeast
- pitch most of the yeast/wort mixture and leave a little in the flask or bottle
- add some more fresh wort to ensure there will be well enough for taking a hydrometer sample later.

If you can't (or don't want to) spare any yeast from the yeast that is pitched (pitching from a vial or pack of dry yeast), you can also use dry bread yeast instead. Bread yeast seems to behave like ale yeasts when it comes to types of sugars that are fermented. I generally find when performing the fast ferment test with the pitched yeast and bread yeast in parallel, that bread yeast shows about the same final extract as ale yeasts and a final extract that is 0.2 – 0.3 Plato higher than a lager yeast. ½ tsp dry bread yeast to 240 ml (8 oz) of wort is plenty. Since slight differences between bread yeast and the actually used yeast seem to exist, it is best to perform the fast ferment test with the yeast that was pitched. Bread yeast however will also show you the final extract as well, but possibly with an error of up to 0.5 Plato or 2 specific gravity points.

Cap the bottle or flask with tin foil or airlock. An airlock is beneficial as the absence of O₂ will keep acetobacter bacteria down and the sample will not taste as sour later. It also makes sure that the sugar will be fermented. But even on a stir plate the amount of aerobic metabolism should be marginal due to the Crabtree effect. Yeast will not metabolize sugars aerobically if their concentration is high enough (> 0.5-1%). I rarely use an airlock these days out of convenience.

Place the test at a warm place (20C / 70F and above, warmer for ale yeasts) and shake it occasionally to keep the yeast in suspension. You may also place it on a stir plate if you have one to spare for that. It should take 1 or 2 days for the fermentation to be over and I tend to give it another 2 days until I see that the sample lies completely flat and no CO₂ escapes when I shake it. When dealing with high gravity worts (18+ Plato, 1.072+ SG), I may also add some air/O₂ to the headspace and dissolve it into the sample through shaking just to kick start any stalled fermentation.

Now you can measure the extract (gravity) of the sample beer. This will give you the lower limit of the final extract (final gravity) that you can expect from this wort. Depending on yeast characteristics and fermentation conditions, the actual beer final extract (final gravity) may be a higher than that. See Understanding Attenuation for details on that. Give it a taste as well. It generally tastes like bad home brew because of the higher fermentation temperatures.

How does this test help me with my brewing?

There are a number of benefits to knowing the limit of attenuation of the wort that is fermented. So many in fact, that I now perform the test for every beer I brew (I used to do the test only for lagers). It became as essential as taking a starting extract (starting gravity) reading or recording the mash temperature.

It will answer an important question when the final extract (final gravity) of the beer is not as expected: Is the problem mashing or fermentation? Based on the answer you can target your efforts of fixing the problem (see Understanding Attenuation for details)

It will also allow you to start lagering or cold conditioning earlier. Now you are not looking for the gravity readings to stabilize, you are looking for a targeted gravity to be reached. Note that your target FG is not necessarily the FG of the fast ferment test, but may be a little higher than that. How much higher depends on the yeast strain and style of beer.

John Palmer [Palmer, 2006] was referring to brewing lagers as flying blind because you don't know when they are done fermenting. With a fast ferment test you will be able to determine the FG of your lager (or ale) before the actual batch has finished fermenting. This is especially helpful for all grain brewers because the final gravity of the beer can greatly be influenced by mashing.

Sometimes, the yeast may stall before reaching the targeted FG (based on the fast ferment test result and yeast performance expectations). In this case you can accept it and fix it with the next batch (better aeration, other yeast strain, temperature) or try to kick start the fermentation by warming it up, rousing the yeast or pitching fresh yeast. I have done the latter when I found that a beer was still too sweet but the yeast didn't want to attenuate further. The knowledge of the limit of attenuation and a measurement of the attenuation of the beer confirmed that the yeast can go further if I help it along.

When cold conditioning before bottle conditioning, the process of cold conditioning can be started as soon as the extract (gravity) falls below a level that leaves enough fermentable sugars in the beer than are needed for carbonation. Here is an example. I recently brewed a Weissbier with a wort that showed a final extract of 2.5 Plato (1.010 SG) in the fast ferment test. The actual fermentation was at 3.5 Plato (1.014 SG) after 5 days. Since I was in a rush to get this beer to cold conditioning, I felt comfortable moving it to the 10C (50F) fridge. Even though this stalled the fermentation and the FG won't be getting much closer to the targeted FG of 2.6 Plato (1.0105 SG), I know that I can account for the amount of fermentable sugars left in the beer when it comes to calculating the amount of sugar needed for bottle conditioning (see [Accurately Calculating Sugar Additions for Carbonation](#) for details on that). But when that level of precision is needed for the gravity readings, I suggest investing in a hydrometer for the range 0.980 – 1.020, which makes the final gravity readings more precise.

Performing a fast ferment test on beer

The idea of this test is to troubleshoot a high finishing gravity without having done a fast ferment test upfront. In the most common form the brewer finds that the fermentation stalled at an unexpectedly high finishing gravity and it should be determined if it makes sense to attempt a restart of the fermentation.

I still have to test if this would work with bread yeast on an actual batch of fermented beer for which I know the limit of attenuation

Appendix

Yeast dependency of the fast ferment test

The following are results from fast ferment tests that were done with different yeasts for different worts

Wort A:

- WY 2206 (Bavarian Lager) FFT AE = 2.2 Plato
- Fleischman Active Dry Bread Yeast FFT AE = 2.4 Plato

Wort B:

- WY 3056 (Bavarian Wheat) FFT AE = 2.6 Plato
- Fleischman Active Dry Bread Yeast FFT AE = 2.8 Plato

Wort C:

- WY 3056 (Bavarian Wheat) FFT AE = 2.5 Plato
- WY 3333 (Bavarian Wheat) FFT AE = 2.6 Plato
- WY 3068 (Bavarian Wheat) FFT AE = 2.6 Plato
- Fleischman Active Dry Bread Yeast (1/4 tsp to 150 ml) FFT AE = 2.5 Plato
- Fleischman Active Dry Bread Yeast (1/2 tsp to 150 ml) FFT AE = 2.5 Plato

This data shows that there can be slight differences in the apparent extract (AE) of the fast ferment tests when done with different yeasts. But these differences are generally small. The best way to do a fast ferment test is to use the yeast that the actual batch of beer is fermented with. In a pinch, bread yeast can also be used.

Note that there was a 0.2 Plato difference between WY3056 and the bread yeast for wort B, while that was not the case for wort C. WY 3056 is a blend and the yeast pitched to wort B was grown from a slant culture (not single cell propagation) and the yeast for wort C was grown from a plate culture (single cell propagation). Maybe that shifted the attenuation capability of the yeast.

Sources

[Narziss, 2005] Prof. Dr. agr. Ludwig Narziss, Prof. Dr.-Ing. habil. Werner Back, Technische Universitaet Muenchen (Fakultaet fuer Brauwesen, Weihenstephan), *Abriss der Bierbrauerei*. WILEY-VCH Verlags GmbH Weinheim Germany, 2005

[Palmer, 2006] John J. Palmer, *How to Brew*, Brewers Publications, Boulder CO, 2006

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